

Ridha Chowdhury

Milwaukee, WI | 571-919-0458 | chowdhuryridha@gmail.com | ridhachowdhury.com | linkedin.com/in/ridha-chowdhury | github.com/RidhaChowdhury

Summary

Software engineer with full-stack development, ML/AI, and team leadership experience. Led teams in development of AI solutions for pharmaceutical and industrial applications. Seeking to apply technical expertise to solve real-world problems.

Education

The Pennsylvania State University, BS in Computer Science | **GPA: 3.20/4.00**

Aug 2021 – May 2025

Experience

Matic | Remote

Nov 2025 – Present

Frontend Engineer

- Led infrastructure migration from AWS Amplify to Cloudflare Workers with Terraform IaC, CI/CD pipelines, and R2 video hosting.
- Drove codebase-wide TypeScript migration from 26% to 89% coverage with strict mode and Jest test infrastructure.
- Built side cart with real-time gift personalization and architected header/navigation redesign with sticky scroll, mobile menu, and keyboard accessibility.

Rockwell Automation | Milwaukee, WI

Jun 2025 – Nov 2025

Software Engineer

- Rebuilt front-end of a plant up-time monitoring application in React. Leveraging ShadCN, Tailwind, and Vite.
- Led a team of 4 in the development and API implementation of an AI Agent designed to generate pharmaceutical recipes with a use case specific interface.

Penn State Research Laboratory | State College, PA

Aug 2024 – May 2025

Lead Product Engineer

- Led cross-functional team of five in product development lifecycle for IoT Point of Care device, acting as product manager and lead engineer with monthly CDC stakeholder presentations.
- Designed and prototyped ESP32-based IoT device using Fusion 360 and additive manufacturing; developed KNN-based ML diagnostic software.

Rockwell Automation | Remote (Milwaukee, WI)

Aug 2024 – May 2025

Part-Time Software Engineer

- Optimized MagneMotion tracking latency by migrating from relational (PostgreSQL) to time-series (InfluxDB) databases.
- Fine-tuned a DBRX LLM on a set of Databricks knowledge bases, enabling Ladder Logic code generation for PLCs.

Rockwell Automation | Milwaukee, WI

May 2024 – Aug 2024

Software Engineering Intern

- Redesigned MagneMotion's feature engineering and classification pipeline using KNN, improving accuracy by 72.6%.
- Deployed MagneMotion's classification pipeline as a microservice into an existing architecture using Docker.
- Evaluated an Apache Spark solution for MagneMotion's against the existing approach interacting with vendor.
- Pioneered on-edge Graph RAG for LLM chatbot web app with HuggingFace Transformers, Neo4J, and LangChain.

Clark Associates | Remote (Lancaster, PA)

Oct 2023 – May 2024

Junior Software Engineer

- Designed and implemented a change log service for four features across front-end and back-end with ASP.NET, SQL, and Blazor. Collaborated with QA, including unit tests and acceptance criteria, for smooth deployment.
- Resolved 120+ SonarQube bugs in ASP.NET back-end, reducing technical debt and improving reliability.

Rockwell Automation | Milwaukee, WI

May 2023 – Aug 2023

Software Engineering Intern

- Built RAG and agentic AI systems using LangChain Retrievers, Agents, and OpenAI embeddings to enhance AI reasoning capabilities.
- Built a Java microservice using Jersey & Hibernate, implementing SQL stored procedures and Azure DevOps CI/CD.

TheCoderSchool | Ashburn, VA

Feb 2020 – May 2023

Software Engineering Instructor

Skills

Languages: Python, TypeScript, Java, C#, C++, HTML+CSS **Frameworks:** React, Next.js, Svelte, ASP.NET Core, Node.js, FastAPI, Flask
Dev Tools: Terraform, Git, GitHub, Azure DevOps, Jira, Agile, SDLC **Systems:** Linux, Docker, Kubernetes, AWS, Azure Cloud, EAS, CUDA
SQL/NoSQL DBs: PostgreSQL, MySQL, SQLite, MongoDB **UI Libraries:** TailwindCSS, Bootstrap, Redux, Axios, Recharts, Three.js, Tanstack
ML/AI: LangChain, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Hugging Face Transformers, KNN/XGBoost, NLTK, OpenAI